

CUSTOMER CHURN | BUSINESS SUMMARY

Which bank customers are about to leave, and why

20.4% of 10,000 customers left. A model can flag the riskiest ones early enough for the bank to act.

Five factors explain who leaves

Number of products	Two products is the sweet spot	7.6% (2) vs 27.7% (1)
Age	Risk peaks between 41 and 60	34% (41-50), 56% (51-60)
Inactivity	Inactive members leave 2x as often	26.9% vs 14.3% active
Account balance	Holding a balance raises risk	24.1% vs 13.8% at zero
Gender	Women leave more; review fairness	25.1% vs 16.5% men

Churn rates from 10,000 customers. Patterns in the data, not proven causes.

The model ranks risk well

0.873

ROC-AUC on customers it
never saw

69%

of flagged customers really leave

59%

of all leavers are caught

One column was removed on purpose. Complaints are recorded after a customer decides to leave (99.8% of leavers), so using them would look perfect and be useless for early warning.

HistGradientBoosting, best of five models. Results on a held-out test set of 1,500 customers.

Offers pay off only above the right risk level

Illustrative economics

An offer costs 100. It saves 30% of at-risk customers. A saved customer is worth 1,000.

Where to draw the line

Flag customers scoring 71% or higher. Net benefit stays flat between about 50% and 80%, so the team can catch more leavers at little cost.

Replace the placeholder costs with real figures before launch.

Threshold chosen on validation data to maximise expected net benefit.

Recommended next steps

Re-engage inactive members

They leave about twice as often. Reach out before they go.

Target ages 41-60 and Germany

They churn at 32-56%, against 20% overall.

Review 3-4 product bundles

326 customers, 83-100% churn. Check how these are sold.

Test before scaling

A/B test offers, remove gender from the model and monitor scores monthly.